

An inverse theorem for extremal generalized Narkiewicz sequences over C_3^3

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Abstract

Let $G = (\mathbb{Z}/3\mathbb{Z})^3$. A short zero-sum witness in an indexed sequence over $G \setminus \{0\}$ is a nonempty set of at most three positions whose entries sum to zero. The generalized Narkiewicz threshold for two such witnesses having a nonzero intersection sum is known to be 25. We prove the complete inverse statement at length 24: every pair of short zero-sum witnesses has zero intersection sum if and only if, up to a permutation of the original positions, the sequence is U^3 , where U consists of eight distinct nonzero elements and has no short zero sum. Thus the known threefold constructions exhaust the extremal sequences. The necessity proof combines the published packing and representative-deletion method with short-free residual structure. Swapping representatives forces constant blocks in the length-16 residual case; the uniquely determined singleton of a length-15 residual excludes the remaining packing case.

Keywords: inverse zero-sum problem; generalized Narkiewicz constant; extremal sequence; positional intersection.

Mathematics Subject Classification (2020): 11B75, 11B30.

1 Introduction

Zero-sum thresholds specify how many terms force a prescribed additive configuration. Their inverse problems ask which sequences avoid that configuration at the last possible length. For intersection conditions on zero-sum subsequences, an inverse theorem must control both multiplicities and the positions of the witnesses: retaining only the support discards part of the hypothesis.

Gao, Hui, Li, Li, Qu and Zhong [3] introduced generalized Narkiewicz constants using pairs of zero-sum subsequences whose intersection has nonzero sum. The intersection is taken over the original indices. For the short-zero-sum invariant they proved

$$\eta^N(C_3^3) = 25 \tag{1}$$

and supplied extremal examples of the form U^3 with $|U| = 8$. The direct threshold and this construction are already published; see [3, Lemma 3.3(2)–(3) and Theorem 3.6(4)]. Here we prove that every extremal sequence has precisely that form.

Write $G = C_3^3 = \mathbb{F}_3^3$. If $S = (a_p)_{p \in P}$ is a sequence indexed by a finite set P , and $I \subseteq P$, put $\sigma_S(I) = \sum_{p \in I} a_p$. Call I a *short zero-sum witness* when $1 \leq |I| \leq 3$ and $\sigma_S(I) = 0$. The empty intersection has sum zero. A sequence is *short-free* if it has no short zero-sum witness, and *squarefree* if its entries are pairwise distinct.

Theorem 1.1. *Let P be any set of cardinality 24, and let $S = (a_p)_{p \in P}$ with $a_p \in G \setminus \{0\}$ for every p . The following are equivalent.*

1. *For every pair $I, J \subseteq P$ of short zero-sum witnesses,*

$$\sigma_S(I \cap J) = 0. \quad (2)$$

2. *There are eight distinct nonzero elements $u_1, \dots, u_8$ such that $U = u_1 \cdots u_8$ is short-free, and a bijection $\phi : \{1, \dots, 8\} \times \{1, 2, 3\} \rightarrow P$ satisfying*

$$a_{\phi(i,j)} = u_i \quad (1 \leq i \leq 8, 1 \leq j \leq 3). \quad (3)$$

Equivalently, $S = U^3$ as a multisequence, or after a permutation of all 24 original positions.

The quantifier in part (1) permits $I = J$, when the condition is automatic. It is therefore the exact negation of having two innerly non-zero-sum-joint short zero-sum subsequences in the terminology of [3]. The theorem does not say that S is short-free: in the conclusion, each three-position fibre is itself a short zero sum.

The substance of the inverse conclusion is that arbitrary multiplicities and possible two-term zero sums cannot survive at length 24. We retain the packing method from the proof of [3, Lemma 3.3(3)], and use the rank-three residual results of Fan, Gao, Wang, Zhong and Zhuang [2]. Equality bookkeeping leaves four packing signatures. Universal choice of deleted representatives then forces eight constant triples in three of these cases and contradicts the fourth. The residual profile and completion argument used below are already present in the proof of [2, Lemma 27(3)]; the contribution here is their use in the complete inverse argument, particularly the interaction between two independently variable packed blocks.

Section 2 gives concrete examples and explains the positional condition. Section 3 obtains the four signatures. Sections 4 and 5 settle the two residual lengths, and Section 6 completes both directions of the theorem. Figures depict finite sets and multiplicities exactly; none is evidence in place of the accompanying proof.

2 Preliminaries and worked examples

We write sequences multiplicatively, even though G is additive. Thus a^3 means three occurrences of a , $v_a(S)$ is its multiplicity, and $\text{supp}(S) = \{a : v_a(S) > 0\}$. The notation Sx^{-1} removes one specified occurrence with value x ; it is not multiplication by a group inverse. We write $\sigma(S)$ for the sum of all entries. Whenever two subsequences are compared, their original position sets are retained.

2.1 Why positions matter

Example 2.1 (One extra copy). Fix any $a \neq 0$. The sequence a^3 has exactly one short zero-sum witness, namely all three positions. In a^4 , the witnesses $I = \{1, 2, 3\}$ and $J = \{2, 3, 4\}$ have intersection sum $2a = -a \neq 0$. Thus every sequence satisfying (2) has $v_a(S) \leq 3$. Both sequences have the same one-element support, so their different behaviour is invisible at the support level.

Example 2.2 (A two-term witness can be allowed). The two-position sequence $a(-a)$ satisfies (2): its only short zero-sum witness is the entire pair. By contrast, for $(a, -a, a)$ the witnesses $\{1, 2\}$ and $\{2, 3\}$ intersect in a single position of value $-a \neq 0$. The first example also shows why the length-24 assumption matters: the intersection condition alone does not exclude two-term witnesses.

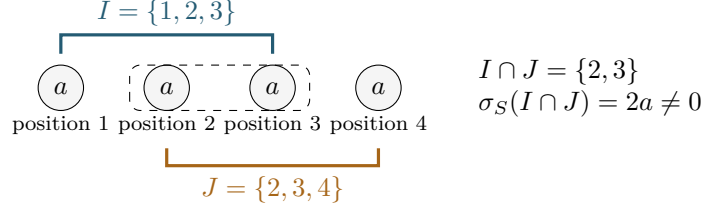


Figure 1: A forbidden positional overlap in a^4 . The two witnesses have the same value-multisquence a^3 but different sets of positions.

2.2 Published inputs

For a finite abelian group H , $\eta(H)$ is the least length forcing a nonempty zero sum of length at most $\exp(H)$, and $\mathfrak{s}(H)$ is the least length forcing a zero sum of length exactly $\exp(H)$. We use s , without the font change, only for the number of packed blocks.

Proposition 2.3 (Published rank-three inputs). *For $G = C_3^3$ the following hold.*

1. $\eta(G) = 17$ and $\mathfrak{s}(G) = 19$.
2. Every short-free sequence of length 16 has total sum zero.

The value of $\eta(G)$ and Property C are recorded in [2, Lemma 7(3),(10)]. The value of $\mathfrak{s}(G)$ follows from [3, Lemma 2.6(4), the preceding Property D discussion, and Lemma 2.10]. Part (2) is [2, Lemma 28]. That paper also proves that a short-free length-15 sequence has nonzero sum [2, Lemma 29(1)]. We shall retain the stronger, explicit singleton identity furnished by its completion argument. These published results are inputs; we do not claim new proofs of their numerical thresholds here.

Lemma 2.4 (Doubling a squarefree core). *If T is a squarefree short-free sequence over $G \setminus \{0\}$, then T^2 is short-free. Consequently $|T| \leq 8$.*

Proof. There is no zero term. A pair with distinct values would already occur in T , whereas a repeated pair has sum $2a \neq 0$. A triple with distinct values would also occur in T . If a zero-sum triple has a repeated value, its sum has the form $2a + b = 0$. In characteristic three this forces $b = a$, but T^2 has only two copies of each value. This proves short-freeness. If $|T| \geq 9$, then $|T^2| \geq 18$, contrary to $\eta(G) = 17$. $\square$

2.3 An explicit extremal sequence

Example 2.5 (An eight-value core). For $(x, y) \in \mathbb{F}_3^2$, let $q(x, y) = x^2 + y^2$, and take

$$U = \prod_{(x,y) \in \mathbb{F}_3^2 \setminus \{(0,0)\}} (x, y, q(x, y)). \quad (4)$$

The values, with all coordinates read modulo 3, are

i	1	2	3	4	5	6	7	8
u_i	(1, 0, 1)	(2, 0, 1)	(0, 1, 1)	(0, 2, 1)	(1, 1, 2)	(1, 2, 2)	(2, 1, 2)	(2, 2, 2).

They are distinct and nonzero. Since $q(v) = 0$ only for $v = 0$ in $\mathbb{F}_3^2$, an antipodal pair of first two coordinates $v, -v$ has third-coordinate sum $2q(v) \neq 0$. For three distinct first-coordinate pairs v, w, z whose sum is zero, $z = -v - w$ and

$$q(v) + q(w) + q(z) = -q(v - w) \neq 0.$$

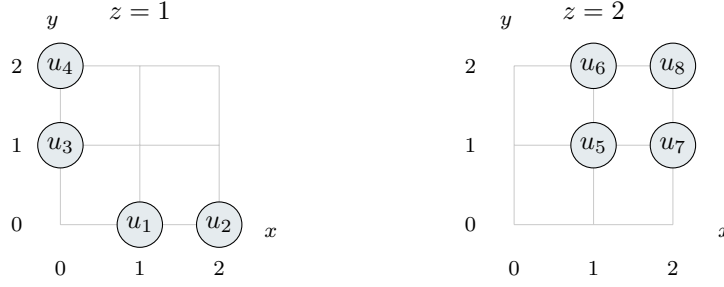


Figure 2: The explicit core in (4), drawn in its two nonempty coordinate layers. The grid records coordinates, not affine incidence; finite-field lines also wrap modulo 3. There are no core points at $z = 0$.

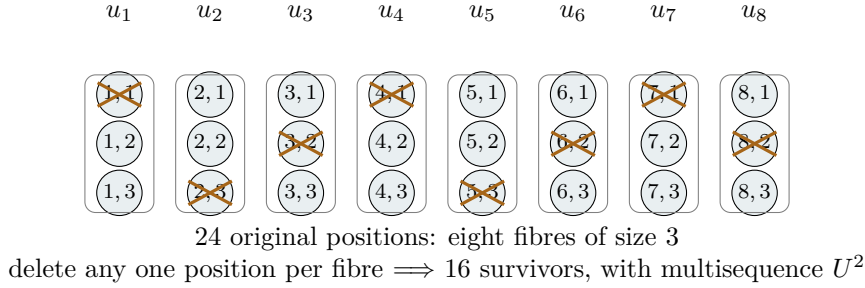


Figure 3: The positional form of U^3 . Every column is one short zero-sum witness. The marked representatives illustrate one of 3^8 possible deletions; every choice leaves the same value multiplicities.

Thus U is short-free. Lemma 2.4 gives a short-free U^2 of length 16, and the sufficiency proof in Section 6 shows that U^3 satisfies (2). This is an illustrative choice within the known construction family.

Example 2.6 (A sharpness check and a length-15 residual). For the explicit U above, $\sigma(U) = 0$: each coordinate sum is divisible by 3. The short-free sequence $R = U^2 u_1^{-1}$ has multiplicities 1, 2, 2, 2, 2, 2, 2 and sum $-u_1 = (2, 0, 2)$. Its unique singleton u_1 is therefore $-\sigma(R)$, illustrating Lemma 5.1 below. The sequence $U^3 u_1$, of length 25, fails (2) by the two overlapping triples on its four u_1 positions, exactly as in Example 2.1.

3 Atom packing and the four signatures

We now prove necessity. Throughout Sections 3–5, $S = (a_p)_{p \in P}$ has 24 nonzero entries and satisfies (2). A nonempty zero-sum sequence is an *atom* if it has no proper nonempty zero-sum subsequence.

Lemma 3.1 (Disjointness of distinct witnesses). *Every short zero-sum witness in a sequence over $G \setminus \{0\}$ is an atom of size 2 or 3. If the sequence satisfies (2), any two distinct short zero-sum witnesses are disjoint.*

Proof. A witness cannot be a singleton. A two-term witness is minimal. A three-term witness cannot contain a zero-sum pair, since its remaining entry would then be zero, so it too is minimal. If witnesses I, J intersect nontrivially, (2) makes their intersection a nonempty zero-sum subset of each. Minimality implies $I = I \cap J = J$. $\square$

Choose a maximal family of pairwise position-disjoint short atoms, and write the corresponding positional decomposition as

$$S = B_1 \cdots B_l B_{l+1} \cdots B_s S_0, \quad |B_i| = \begin{cases} 3 & i \leq l, \\ 2 & i > l. \end{cases} \quad (5)$$

Here $r = |S_0|$, and maximality means that the remaining positions S_0 contain no short zero sum. Such a packing exists because P is finite. In fact Lemma 3.1 shows that every short witness is a packed block: otherwise it would be disjoint from all the blocks and could be added. This observation makes the universal choice in the following published deletion method explicit.

Lemma 3.2 (Universal representative deletion). *For every choice of a position p_i in each B_i , write $x_i = a_{p_i}$. Then*

$$R_x = S(x_1 \cdots x_s)^{-1} \quad \text{is short-free,} \quad (6)$$

$$Q_x = S(x_1 \cdots x_l)^{-1} \quad \text{has no three-term zero sum.} \quad (7)$$

Proof. Any short witness in R_x would be a short witness in S , hence would be one of the B_i . But each B_i has lost a position. Likewise a three-term witness in Q_x would be one of $B_1, \dots, B_l$, each of which has lost a position. The argument did not restrict the selected representatives. $\square$

Lemma 3.2 is the representative-deletion argument from the proof of [3, Lemma 3.3(3)], applied before using any threshold length assumption. Its universal quantifier is essential: the proof below compares different transversals of the same fixed packing.

Proposition 3.3 (Four possibilities). *Every packing (5) has one of the signatures*

$$(l, s, r) \in \{(6, 8, 2), (7, 8, 1), (8, 8, 0), (6, 9, 0)\}. \quad (8)$$

Proof. By Proposition 2.3 and Lemma 3.2,

$$24 - l = |Q_x| \leq 18, \quad 24 - s = |R_x| \leq 16.$$

Thus $l \geq 6$ and $s \geq 8$. Length accounting gives

$$24 = 3l + 2(s - l) + r = l + 2s + r, \quad l \leq s, \quad r \geq 0. \quad (9)$$

It follows that $s \leq 9$. If $s = 8$, then $l + r = 8$, so $(l, r) = (6, 2), (7, 1)$ or $(8, 0)$. If $s = 9$, then $l + r = 6$; hence $(l, r) = (6, 0)$. $\square$

4 Length-16 survivors force constant blocks

Proposition 4.1. *If $s = 8$, then $(l, s, r) = (8, 8, 0)$ and every B_i is a constant three-term sequence. The sequence has the form U^3 for a squarefree short-free U of length 8.*

Proof. For every transversal x , Lemma 3.2 gives a short-free R_x of length 16. Proposition 2.3(2) therefore gives

$$\sigma(S) - \sum_{i=1}^8 x_i = 0. \quad (10)$$

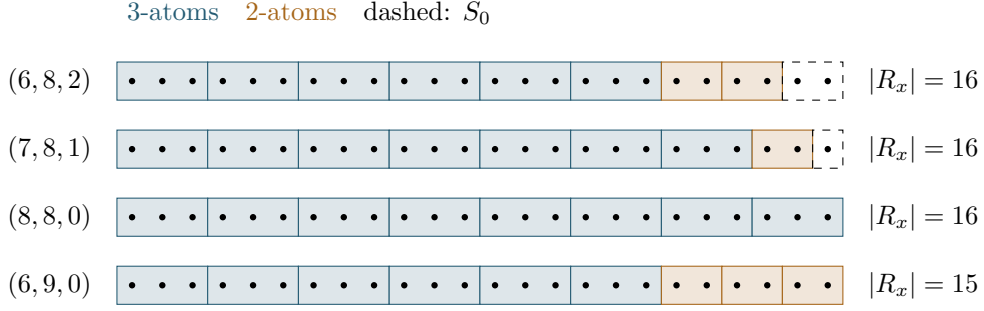


Figure 4: All four length budgets. Each dot is an original position; each solid box is a zero-sum block. These are candidate signatures under the hypothesis, not asserted examples. Only the third row survives.

Fix a block B_j and fix the representative positions in all other blocks. Let t be the sum of their values. If two positions in B_j have values a, b , applying (10) with each choice gives

$$\sigma(S) - t - a = 0 = \sigma(S) - t - b.$$

Hence $a = b$. All entries within every packed block are equal.

A constant two-term block would have sum $2a = 0$, which forces $a = 0$ in G , contrary to the alphabet. Thus there are no two-term blocks. The first two signatures in (8) are excluded, leaving $(8, 8, 0)$. Write $B_i = u_i^3$ and $U = u_1 \cdots u_8$. Deleting any one position from each block leaves U^2 , which is short-free. If $u_i = u_j$ for $i \neq j$, then U^2 would contain at least four copies of this value and hence a zero-sum triple. Therefore U is squarefree. Being a subsequence of U^2 , it is also short-free. $\square$

Example 4.2 (What the representative swap compares). If a hypothetical two-term packed block has values $a, -a$, keep all other deletions fixed and denote the other 15 survivors by W . The two residuals are $W(-a)$ and Wa . When $s = 8$, both have length 16, so both must have sum zero. Their sum difference is $2a \neq 0$, which is impossible. For a three-term block the same comparison forces any two selectable values to agree. The comparison concerns two deletions of one sequence, without assuming that a preferred deletion represents every transversal.

5 Length-15 survivors exclude nine blocks

5.1 The unique singleton

We record the precise residual fact needed for exchanges. Its multiplicity and completion reasoning follows [2, proof of Lemma 27(3), together with Lemma 28]. We include the details to identify the singleton in each positional residual.

Lemma 5.1. *Every short-free length-15 sequence R over $G \setminus \{0\}$ has exactly eight support values, seven of multiplicity two and one of multiplicity one. If u is the singleton value, then*

$$u = -\sigma(R). \tag{11}$$

In particular $\sigma(R) \neq 0$.

Proof. Every multiplicity is at most two, since a^3 has sum zero. Thus at least eight values occur. Choosing one copy of every support value produces a squarefree short-free sequence, whose length is at most eight by Lemma 2.4. Exactly eight values occur, and their positive multiplicities sum to 15, giving the profile $2^7 1$.

Let T be the squarefree support sequence. Appending the singleton u gives $Ru = T^2$, which is short-free by Lemma 2.4 and has length 16. Proposition 2.3(2) now yields $0 = \sigma(Ru) = \sigma(R) + u$. Finally $u \neq 0$. $\square$

5.2 The two possible multiplicity exchanges

Suppose the last signature in (8), $(l, s, r) = (6, 9, 0)$, occurs. All positions belong to zero-sum blocks, so $\sigma(S) = 0$. For every transversal $x = (x_1, \dots, x_9)$, the residual R_x is short-free of length 15, with unique singleton value

$$u_x = -\sigma(R_x) = \sum_{i=1}^9 x_i. \quad (12)$$

Fix a block B_j with two distinct selectable values $a \neq b$. Fix representatives in the other eight blocks, with value-sum t . Let R_a and R_b be the residuals obtained by deleting the chosen position of value a and of value b , respectively. The choices are distinct positions in the same block. As multisets of values,

$$R_b = R_a a b^{-1}. \quad (13)$$

The removed copy of b is present in R_a , so this equation is legitimate.

Lemma 5.2. *For fixed complementary representatives as above, the counts $(v_a(R_a), v_b(R_a))$ and $(v_a(R_b), v_b(R_b))$ must follow exactly one of the transitions*

$$(0, 1) \mapsto (1, 0), \quad (1, 2) \mapsto (2, 1). \quad (14)$$

The first transition is impossible under (12); the second forces $t = 0$.

Proof. By (13), the count of a increases by one and that of b decreases by one. All counts in each residual lie in $\{0, 1, 2\}$, and exactly one value has count one. Initially the count of a is 0 or 1, and that of b is 1 or 2. The possibility $(1, 1)$ already gives two singletons. The possibility $(0, 2)$ becomes $(1, 1)$ and gives two singletons after the exchange. The two remaining possibilities are (14).

In the first case the singleton of R_a is b , and that of R_b is a . Equation (12) gives

$$b = t + a, \quad a = t + b.$$

Subtracting these equations gives $2(a - b) = 0$, hence $a = b$ since multiplication by two is invertible in G . This is a contradiction.

In the second case the singletons of R_a, R_b are a, b , respectively. Their identities give

$$a = t + a, \quad b = t + b,$$

and therefore $t = 0$. $\square$

Proposition 5.3. *The signature $(6, 9, 0)$ cannot occur.*

Proof. It has three two-term zero-sum blocks. Each has distinct values $a, -a$: if they were equal, $2a = 0$ would contradict $a \neq 0$. Choose one as B_j . For *every* choice of representatives in the other eight blocks, Lemma 5.2 forces their sum t to be zero. Choose a second two-term block B_k , $k \neq j$, whose distinct values are c, d . Keep the remaining seven representatives fixed and let their sum be h . Choosing c and then d in B_k gives

$$h + c = 0 = h + d.$$

Thus $c = d$, a contradiction. $\square$

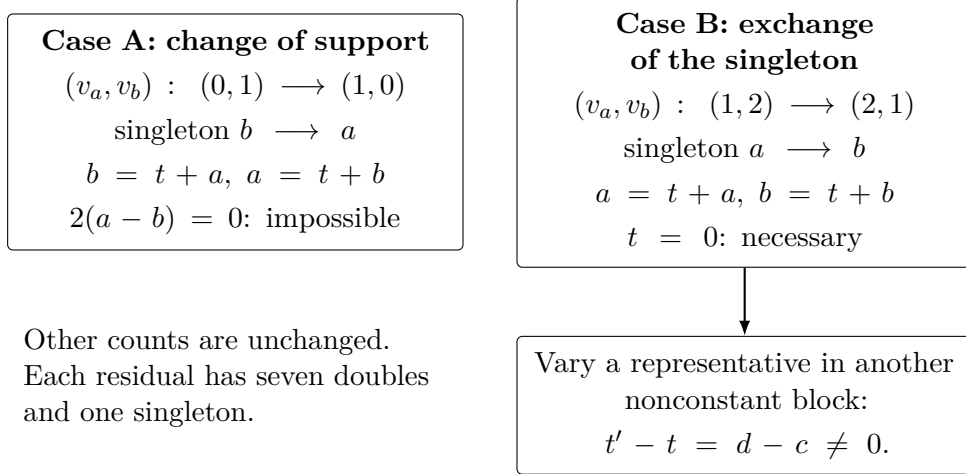


Figure 5: The exhaustive exchange in a hypothetical nine-block packing. Case A contradicts $a \neq b$. Case B requires zero complementary sum for every choice, which another nonconstant block makes impossible.

Example 5.4 (The profile exchange is not itself a contradiction). For distinct a, b , the abstract profiles $ab^2c_1^2 \cdots c_6^2$ and $a^2bc_1^2 \cdots c_6^2$ both have length 15 and shape 2^71 . They illustrate the second transition; they are not asserted to be short-free sequences. The obstruction uses the singleton *sums* as well as the counts, and then varies a representative in a different block. This separates the local multiplicity comparison from the global nine-block contradiction.

6 Completion of the classification

Proof of Theorem 1.1. Assume (1). Proposition 3.3 supplies the four signatures. Proposition 5.3 excludes $s = 9$; Proposition 4.1 then leaves eight constant three-position blocks with distinct values $u_1, \dots, u_8$, and their squarefree core U is short-free. Number the three original positions in each block arbitrarily. Sending (i, j) to the j th position of block i defines the bijection ϕ in (3). This explicitly accounts for all 24 original positions and proves (2).

Conversely, assume (2), and work first on the product index set $\{1, \dots, 8\} \times \{1, 2, 3\}$. There is no singleton witness. A pair with different values would be a short zero sum in U , and a pair with equal value a has sum $2a \neq 0$. In a zero-sum triple, three distinct values would again give a short zero sum in U . If a value a repeats, $2a + b = 0$ implies $b = a$. The witness is therefore exactly the full three-position fibre of a , since there are only three positions with that value.

Consequently any two short witnesses are either identical fibres, whose intersection has sum $3a = 0$, or different fibres, whose intersection is empty. The bijection ϕ preserves intersections and all entry sums: for position sets I, J , $\phi^{-1}(I \cap J) = \phi^{-1}(I) \cap \phi^{-1}(J)$. Transporting the conclusion proves (2) on the original P . This is a direct positional verification of the already known construction from [3, proof of Lemma 3.3(2)]. $\square$

Corollary 6.1. *Every length-24 sequence satisfying (2) has exactly eight short zero-sum witnesses. They are disjoint constant triples and partition the original position set. Every support value has multiplicity three.*

Proof. The last part of the proof of Theorem 1.1 identifies all witnesses as the eight full value fibres. Each such fibre is a witness. $\square$

7 Discussion and formal verification

7.1 The inverse content

Theorem 1.1 complements the published threshold by determining every multiplicity and every short witness in every extremal sequence. At residual length 16, fixed total sum forces equal entries within blocks. At length 15, the singleton identity couples independently variable representatives, excluding the competing packing. These deductions supply the necessity beyond the known construction. The argument uses specific rank-three residual facts and no enumeration of extremal sequences. We make no assertion for other ranks, exponents or lengths. Nor do we claim a new core classification: the core normal form in [2, Lemma 9] is prior work.

7.2 A geometric interpretation of the core

After the multiplicity conclusion has been proved, the core may be viewed as a set of eight points. Adjoining 0 gives a nine-point cap in $AG(3, 3)$: three distinct points in characteristic three are collinear exactly when their sum is zero. A line through 0 corresponds to a forbidden pair $a, -a$ in U , and a line not through 0 corresponds to a forbidden distinct triple in U . This explains the core in Figure 2; it does not supply the necessity of the threefold multiplicities on the original positions.

7.3 Formal statement and provenance

The positional equivalence of Theorem 1.1 has a Lean formalization, registered as PALOMAR-2026-10-09-000011, version 1 [1]. The declaration is:

`InverseZeroSum.Candidate3.frozenClassification`

It uses $\text{Fin}(24)$ positions, nonzero entries, finite position sets for witnesses, and an explicit permutation for threefold repetition. Both implications and the required source-input propositions are proved. The registration supplies the immutable source revision, using Lean 4.35.0-rc2. Formal registration concerns the encoded theorem; the exposition and source attributions above stand separately.

AI assistance. OpenAI ChatGPT was used substantively in developing the mathematical arguments, constructing the Lean formalization, checking literature and preparing the manuscript, examples and diagrams. Responsibility for the article and its attributions remains with the author.

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